

ECE501 - DIGITAL IMAGE PROCESSING

SAR-Optical Feature Fusion for Crop-Type Classification REPORT - WEEK 2

Name	Enrollment No.
Gamit Srushti	AU2440205
Chavda Dhruvi	AU2440236
Chaithanya J	AU2440097

GitHub repository:

https://github.com/srushtibeng-svg/DIP_2026_Project19-SOFF_Group20.git

Assigned Data: Germany + Uruguay (Group A), YieldSAT Sentinel-2 + crop labels + yield masks; Sentinel-1 SAR acquired by the group via Google Earth Engine.

1) Work done this week

→ **Dataset study** from the YieldSAT documentation we studied the data layout: Sentinel-2 L2A time series at 10m (about 5-day revisit), 13 spectral bands plus the Scene Classification Layer (SCL) for cloud masking, and rasterised per-pixel yield maps. We selected the bands needed for our features: B02 (blue), B04 (red), B05 (red edge 1), B08 (NIR) and B11(SWIR 1).

→ **Google Earth Engine.** Registered a non-commercial Earth Engine project and connected it to Google Colab through the Python API. As a first test we queried Sentinel-1 GRD (IW mode, VV and VH) for one point in Germany.

→ **Feature functions.** Wrote functions for optical indices (NDVI, NDRE, NDMI, EVI), GLCM texture (contrast, homogeneity, energy, correlation), SAR backscatter statistics (VV and VH mean and variance, VV/VH ratio) and feature concatenation. These were checked on synthetic arrays only and have not yet been run on real YieldSAT patches.

Sentinel-1 acquisition test (Google Earth Engine)

Parameter	Value
Location	Single test point in Germany
Period	April - September 2021 (growing season)
Product / mode	Sentinel-1 GRD, Interferometric Wide (IW)
Polarisations required	VV and VH
Image found	116
Bands returned	VV, VH, angles

2) Assigned Dataset Summary

→ Crop and field counts for our country pair, as listed in the YieldSAT documentation:

Country	Crop	Fields	Avg.field	Years
Germany	Rapeseed	111	21.6(country avg.)	2016-2022
Germany	Wheat	188	21.6(country avg.)	2016-2022

Uruguay	Soybean	572	57.3(country avg.)	2018-2022
---------	---------	-----	--------------------	-----------

3) Work Planned for Next Week

Data. We will inspect the folder and file structure, check the band order and label format, and load sample patches for Germany and Uruguay. For each sample we will visualise the RGB composite, the NDVI image and the yield mask, and use the Scene Classification Layer to check how much cloud cover affects the patches. The data-loading script in our repository, which is still a scaffold written for an assumed format, will then be adapted to the real files.

Optical and SAR features. With real patches loaded, we will compute the optical features (NDVI, NDRE, EVI, and GLCM texture) and save the first result figures to the Results folder. In parallel, we will extract field coordinates and acquisition dates from the YieldSAT metadata and request the matching Sentinel-1 data from Google Earth Engine. Because the free Earth Engine tier has a limited compute quota, this will be done for a small set of fields in batches instead of whole regions. A Lee filter will be applied to reduce speckle, after which we will compute the SAR features: mean and variance of VV, the VV/VH ratio and GLCM texture.

Classification target and baseline. We will train a first baseline Random Forest on a small sample of fields. The train and test split will be done by field rather than by pixel, so that pixels of the same field never appear in both sets and the accuracy is not inflated. These first results will be preliminary and will only be used to check that the pipeline works end to end.

4) Literature Review

- **Miranda et al., YieldSAT (CVPR 2026).** Our primary dataset: 2,173 fields in four countries with Sentinel-2 time series and pixel-level yield maps. It contains no SAR data, hence the separate Sentinel-1 acquisition.
- **Mena et al., adaptive fusion of multi-modal remote sensing data for sub-field crop yield prediction (Remote Sensing of Environment, 2025).** Fusion on the same benchmark using deep learning; a reference point for why fusion helps, while we use classical features.

- **Blickensdorfer et al., crop-type and crop-sequence mapping for Germany from combined Sentinel-1, Sentinel-2 and Landsat 8 time series (Remote Sensing of Environment, 2022).** Relevant to our German fields; multi-sensor time series for crop mapping.
- **Van Tricht et al., textural features for image classification (IEEE Trans. SMC, 1973).** Origin of the GLCM texture features.
- **Lee, digital image enhancement and noise filtering by use of local statistics (IEEE TPAMI, 1980).** Basis of the Lee filter planned for SAR speckle reduction.
- **Breiman, Random Forests (Machine Learning, 2001).** Classifier for the baseline models.
- **Gorelick et al., Google Earth Engine (Remote Sensing of Environment, 2017).** Platform used for Sentinel-1 acquisition.

5) Challenges Faced

Crop type versus country (correction to week 1). Our Week 1 report stated that the crops are present in both countries. After studying the dataset documentation this week, we found that Germany contains only rapeseed (111 fields) and wheat (188 fields), while Uruguay contains only soybean (572 fields); no crop occurs in both countries. This matters because a single three-class classifier could learn to separate the countries, which differ in climate, season and imaging conditions, instead of learning the crops themselves, and its accuracy would then look better than it really is. We are considering two options: (A) classifying rapeseed versus wheat within Germany only, which is a fair two-class problem, or (B) the three-class problem with this limitation stated clearly in the report.

Earth Engine setup and quota. Using Sentinel-1 through Google Earth Engine required creating a Cloud project and completing the non-commercial registration, which took part of the week. The free community tier also has a limited compute quota, so SAR data cannot be requested for all fields at once. We will request it field by field in small batches and export only what is needed.